

2.1 Double Integral

An integrable function of two or three variables defined on the certain interval is called multiple integral. Cauchy introduced the concept of double integrals over rectangles, and Riemann developed a rigorous definition of multiple integrals over general domains in 19th century. During early 20th century, Lebesgue refined the theory of multiple integral introducing the Lebesgue integral which provided a more general framework for integration. Today, multiple integrals are a fundamental tool in mathematics, physics, engineering, and various other fields. They are used to compute quantities such as volume, mass, center of mass, moment of inertia and probability. In this chapter, we define double integrals and triple integrals of functions of two and three variables.

Evaluation of double integral

In a double integral, we integrate a function $F(x, y)$ of the independent variables x and y over a closed and bounded region R in the xy - plane. On division of region R into rectangles by drawing parallel to x and y axes with n elementary areas as

$$\Delta A_1, \Delta A_2, \dots, \Delta A_n.$$

Choose a point (x_i, y_i) within i^{th} rectangles with area ΔA_i as in Fig. 2.1(a), then the sum

$$F(x_1, y_1) \Delta A_1 + F(x_2, y_2) \Delta A_2 + \dots + F(x_i, y_i) \Delta A_i + \dots + F(x_n, y_n) \Delta A_n = \sum_{i=1}^n F(x_i, y_i) \Delta A_i.$$

If this limit exists, as the length of the maximum diagonal of the rectangles approaches to zero when n approaches to infinity, then the limit of this sum is called double integral of $F(x, y)$ over the region R and is denoted by

$$\iint_R F(x, y) dA = \lim_{\substack{n \rightarrow \infty \\ \Delta A \rightarrow 0}} \sum_{i=1}^n F(x_i, y_i),$$

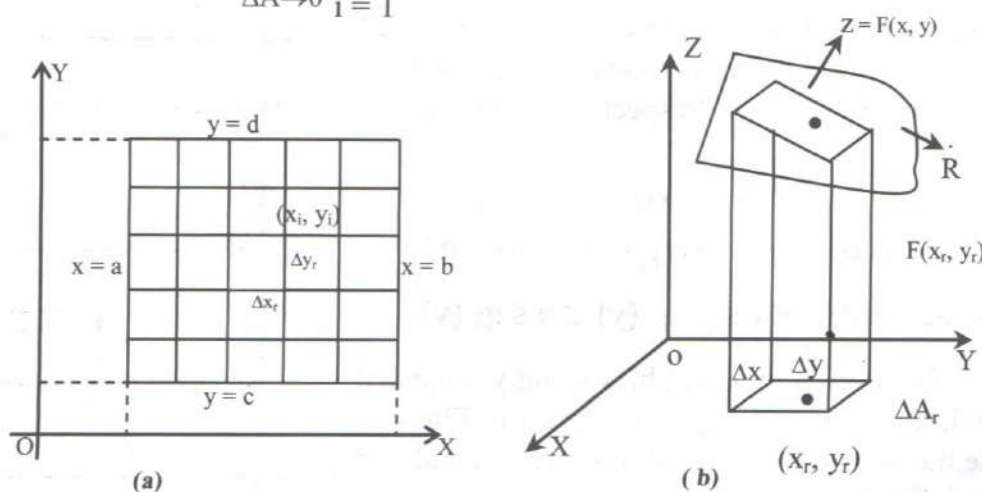


Figure 2.1 Double integral of a function as limit of a sum in (a) and the integral of $z = F(x, y)$ in (b).

provided the limit exists. This can also be expressed as

$$\iint_R F(x, y) \, dx \, dy \quad \text{or} \quad \iint_R F(x, y) \, dy \, dx.$$

2.1.1 Double Integrals in Cartesian Coordinates

The graph of the function of two variables $F(x, y)$ is a surface plotted in three dimension. The double integral of $F(x, y)$ over the region R is the volume between the graph and the region R . Thus, the double integral of $F(x, y)$ is the volume defined as

$$\iint_R F(x, y) \, dx \, dy$$

If $F(x, y) = 1$, then the numerical value of the volume is equal to value of the area of the region R as in shown in Fig. 2.2. Figure 2.2 shows that the area of the region R as given by the formula as

$$A = \iint_R dx \, dy.$$

(a) When the region R is rectangle: $a \leq x \leq b$, $c \leq y \leq d$

Let $F(x, y)$ be a function of two variables x and y , bounded by $x = a$, $x = b$, $y = c$ and $y = d$ as shown in Fig. 2.3. Then, the double integral of $F(x, y)$ with respect to x and y in the region R is defined as

$$\begin{aligned} \iint_R F(x, y) \, dx \, dy &= \int_a^b \left(\int_c^d F(x, y) \, dy \right) dx \\ &= \int_c^d \left(\int_a^b F(x, y) \, dx \right) dy. \end{aligned}$$

(b) When the region R : $a \leq x \leq b$, $f_1(x) \leq y \leq f_2(x)$

Let $F(x, y)$ be a function of two variables x and y , bounded by $x = a$, $x = b$; $y = f_1(x)$, $y = f_2(x)$ as shown in Fig. 2.4, then the double integral of $F(x, y)$ with respect to x and y in the region R is defined as

$$\iint_R F(x, y) \, dx \, dy = \int_a^b \left(\int_{f_1(x)}^{f_2(x)} F(x, y) \, dy \right) dx.$$

(c) When the region R : $c \leq y \leq d$, $g_1(y) \leq x \leq g_2(y)$

Let $F(x, y)$ be a function of two variables x and y , bounded by $y = c$, $y = d$; $x = g_1(y)$, $x = g_2(y)$ as shown in Fig. 2.5, then the double integral of $F(x, y)$ with respect to x and y in the region R is defined as

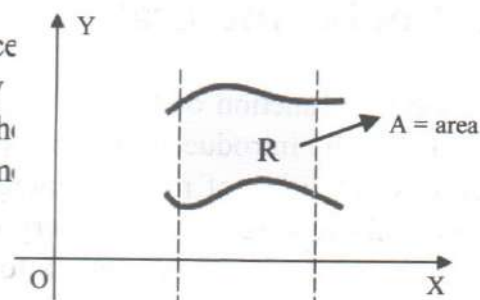


Figure 2.2 Area of the region R when $F(x, y) = 1$.

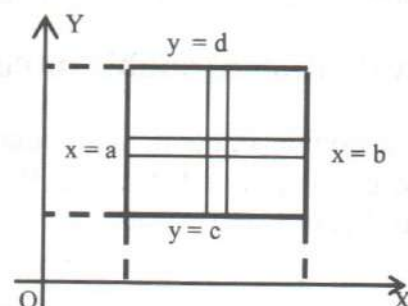


Figure 2.3 Double integral of $F(x, y)$ bounded by the rectangle.

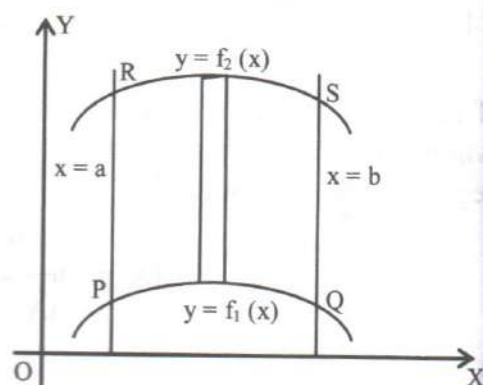


Figure 2.4 Double integral of $F(x, y)$ bounded by $f_1(x) \leq y \leq f_2(x)$, $a \leq x \leq b$.

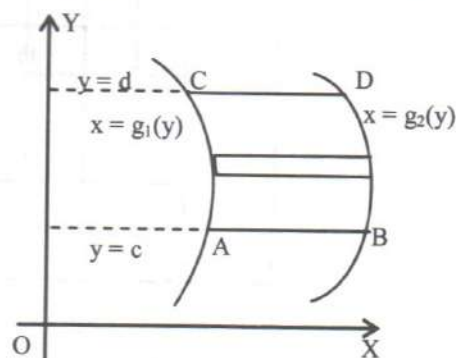


Figure 2.5 Double integral of $F(x, y)$ bounded by $f_1(y) \leq x \leq g_2(y)$ and $a \leq y \leq b$.

$$\iint_R F(x, y) dx dy = \int_c^d \left(\int_{g_1(y)}^{g_2(y)} F(x, y) dx \right) dy.$$

2.1.2 Double Integrals in Polar Coordinates

If a function of two variables $F(r, \theta)$ is bounded by $\theta = \alpha$, $\theta = \beta$, $r = f_1(\theta)$, $r = f_2(\theta)$, then the double integral of $F(r, \theta)$ over the region R is defined as

$$\iint_R F(r, \theta) dr d\theta = \int_\beta^\alpha \left(\int_{f_1(\theta)}^{f_2(\theta)} F(r, \theta) dr \right) d\theta.$$

To evaluate this integral, we first integrate with respect to r between limits $r = f_1(\theta)$ and $r = f_2(\theta)$, keeping θ fixed. The resulting function is integrated with respect to θ from $\theta = \alpha$ to $\theta = \beta$, where α and β are constants.

Figure 2.6 shows that the arcs AB and CD are represented by the curves $r = f_1(\theta)$ and $r = f_2(\theta)$, respectively, bounded by the lines $\theta = \alpha$ and $\theta = \beta$. Let PQ be a small increment of θ with angular thickness $\Delta\theta$. Then

$$\int_\alpha^\beta \left(\int_{f_1(\theta)}^{f_2(\theta)} F(r, \theta) dr \right) d\theta$$

indicates that the integration is performed along PQ and the integration with respect to θ represents the rotation of the strip PQ from AC to BD as shown in Fig. 2.6.

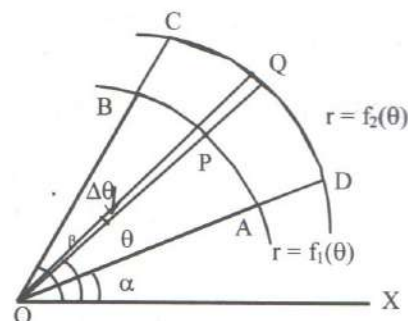


Figure 2.6 Double integral of $F(r, \theta)$ bounded by $r_1 = f_1(\theta)$, $r_2 = f_2(\theta)$, $\theta = \alpha$ to $\theta = \beta$.

Worked out Examples

1. Evaluate $\int_0^1 \int_1^2 (x^2 + y^2) dx dy$

Solution: Here the integral is given by

$$\begin{aligned} \int_0^1 \int_1^2 (x^2 + y^2) dx dy &= \int_0^1 \left(\int_1^2 (x^2 + y^2) dx \right) dy = \int_0^1 \left[\frac{x^3}{3} + y^2 x \right]_1^2 dy \\ &= \int_0^1 \left[\frac{8}{3} + 2y^2 - \frac{1}{3} - y^2 \right] dy = \int_0^1 \left[\frac{7}{3} + y^2 \right] dy = \left[\frac{7y}{3} + \frac{y^3}{3} \right]_0^1 = \frac{8}{3}. \end{aligned}$$

2. Show that $\int_0^1 dx \int_0^1 \frac{x-y}{(x+y)^3} dy \neq \int_0^1 dy \int_0^1 \frac{x-y}{(x+y)^3} dx$

Solution: Here the integral is given by

$$\begin{aligned} \int_0^1 dx \int_0^1 \frac{x-y}{(x+y)^3} dy &= \int_0^1 dx \int_0^1 \frac{-(x+y) + 2x}{(x+y)^3} dy = \int_0^1 dx \int_0^1 \left[\frac{-(x+y)}{(x+y)^3} + \frac{2x}{(x+y)^3} \right] dy \\ &= \int_0^1 dx \int_0^1 \left[-\frac{1}{(x+y)^2} + \frac{2x}{(x+y)^3} \right] dy = \int_0^1 \left[\frac{1}{x+y} - \frac{x}{(x+y)^2} \right]_0^1 dx \end{aligned}$$

$$\begin{aligned}
 &= \int_0^1 \left[\frac{1}{x+1} - \frac{x}{(x+1)^2} - \frac{1}{x} + \frac{x}{(x+0)^2} \right] dx = \int_0^1 \left[\frac{1}{x+1} - \frac{x+1-1}{(x+1)^2} - \frac{1}{x} + \frac{1}{x} \right] dx \\
 &= \int_0^1 \left[\frac{1}{x+1} - \frac{1}{x+1} + \frac{1}{(x+1)^2} \right] dx = \int_0^1 \frac{1}{(x+1)^2} dx \\
 &= - \left[\frac{1}{x+1} \right]_0^1 = - \left[\frac{1}{2} - 1 \right] = \frac{1}{2}.
 \end{aligned}$$

And the integral on right hand side is

$$\begin{aligned}
 \int_0^1 dy \int_0^1 \frac{x-y}{(x+y)^3} dx &= \int_0^1 dy \int_0^1 \frac{(x+y)-2y}{(x+y)^3} dx = \int_0^1 dy \int_0^1 \left[\frac{x+y}{(x+y)^3} - \frac{2y}{(x+y)^3} \right] dx \\
 &= \int_0^1 dy \int_0^1 \left[\frac{1}{(x+y)^2} - \frac{2y}{(x+y)^3} \right] dx = \int_0^1 \left[-\frac{1}{x+y} + \frac{y}{(x+y)^2} \right]_0^1 dy \\
 &= \int_0^1 \left[-\frac{1}{y+1} + \frac{y}{(y+1)^2} + \frac{1}{y} - \frac{y}{(0+y)^2} \right] dy \\
 &= \int_0^1 \left[-\frac{1}{y+1} + \frac{y+1-1}{(y+1)^2} + \frac{1}{y} - \frac{1}{y} \right] dy \\
 &= \int_0^1 \left[-\frac{1}{y+1} + \frac{1}{y+1} - \frac{1}{(y+1)^2} \right] dy = - \int_0^1 \frac{1}{(y+1)^2} dy \\
 &= \left[\frac{1}{y+1} \right]_0^1 = \left[\frac{1}{2} - 1 \right] = -\frac{1}{2}.
 \end{aligned}$$

Hence, $\int_0^1 dx \int_0^1 \frac{x-y}{(x+y)^3} dy \neq \int_0^1 dy \int_0^1 \frac{x-y}{(x+y)^3} dx.$

3. Evaluate $\int_1^2 \int_0^x \frac{dy dx}{y^2+x^2}$

Solution: Here the integral is

$$\begin{aligned}
 \int_1^2 \int_0^x \frac{dy dx}{y^2+x^2} &= \int_1^2 \int_0^x \left[\frac{dy}{y^2+x^2} \right] dx = \int_1^2 \left[\frac{1}{x} \tan^{-1} \frac{y}{x} \right]_0^x dx \\
 &= \int_1^2 \frac{1}{x} \left[\frac{\pi}{4} - 0 \right] dx = \frac{\pi}{4} \int_1^2 \frac{1}{x} dx = \frac{\pi}{4} [\log x]_1^2 = \frac{\pi}{4} \log 2.
 \end{aligned}$$

4. Evaluate $\int_0^1 \int_0^{\sqrt{1+x^2}} \frac{1}{1+x^2+y^2} dx dy$

Solution: Here the integral is

$$\begin{aligned}\int_0^1 \int_0^{\sqrt{1+x^2}} \frac{dx dy}{1+x^2+y^2} &= \int_0^1 \int_0^{\sqrt{1+x^2}} \left(\frac{dy}{1+x^2+y^2} \right) dx = \int_0^1 \left[\frac{1}{\sqrt{1+x^2}} \tan^{-1} \frac{y}{\sqrt{1+x^2}} \right]_0^{\sqrt{1+x^2}} dx \\ &= \int_0^1 \frac{1}{\sqrt{1+x^2}} \frac{\pi}{4} dx = \frac{\pi}{4} [\log(x + \sqrt{1+x^2})]_0^1 = \frac{\pi}{4} \log(1 + \sqrt{2}).\end{aligned}$$

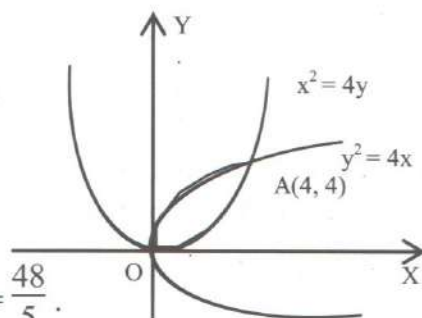
5. Evaluate $\iint_R y \, dy \, dx$ where R is the region bounded by the parabolas $y^2 = 4x$ and $x^2 = 4y$

Solution: Here the integral is

$$\iint_R y \, dy \, dx = \iint_R (y \, dy) \, dx,$$

where R is region bounded by the parabolas $y^2 = 4x$ and $x^2 = 4y$ as in the figure. This is first integrated with respect to y along vertical strip in which y extends from $y = x^2/4$ to $y = 2\sqrt{x}$ and those of x extends from $x = 0$ to $x = 4$. So

$$\begin{aligned}\iint_R y \, dy \, dx &= \int_0^4 \int_{x^2/4}^{2\sqrt{x}} y \, dy \, dx = \int_0^4 \left[\frac{y^2}{2} \right]_{x^2/4}^{2\sqrt{x}} dx \\ &= \frac{1}{2} \int_0^4 \left(4x - \frac{x^4}{16} \right) dx = \frac{1}{2} \left[2x^2 - \frac{x^5}{80} \right]_0^4 \\ &= \frac{1}{2} \left[32 - \frac{16 \times 64}{80} \right] = \frac{1}{2} \left[32 - \frac{64}{5} \right] = \frac{48}{5}.\end{aligned}$$



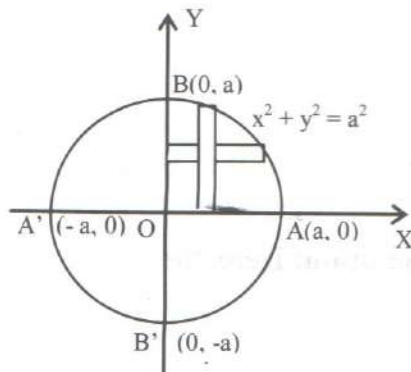
6. Evaluate $\iint_R xy \, dx \, dy$ where R is the quadrant of the circle $x^2 + y^2 = a^2$ and $x \geq 0, y \geq 0$

Solution: Here, the integral is

$$\iint_R xy \, dx \, dy,$$

where R is the region of positive quadrant of the circle $x^2 + y^2 = a^2$ as in the figure in which the integrand is first integrated with respect to x along the horizontal strip which extends from $x = 0$ to $x = \sqrt{a^2 - y^2}$ and those of y varies from $y = 0$ to $y = a$. So

$$\begin{aligned}\iint_R xy \, dx \, dy &= \int_0^a \left(\int_{x=0}^{\sqrt{a^2-y^2}} x \, dx \right) y \, dy \\ &= \int_0^a \left[\frac{x^2}{2} \right]_0^{\sqrt{a^2-y^2}} y \, dy = \int_0^a \frac{(a^2 - y^2)y}{2} dy \\ &= \frac{1}{2} \left[\frac{a^2 y^2}{2} - \frac{y^4}{4} \right]_0^a = \frac{1}{2} \left[\frac{a^4}{2} - \frac{a^4}{4} \right] = \frac{a^4}{8}.\end{aligned}$$



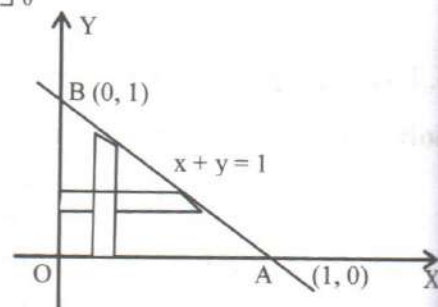
7. Evaluate $\iint_R (x^2 + y^2) \, dx \, dy$ over the region bounded by $x = 0, y = 0, x + y = 1$

Solution: Here the integral is given by

$$\iint_R (x^2 + y^2) \, dx \, dy,$$

where R is the region bounded by $x = 0$, $y = 0$, and $x + y = 1$ as in the figure. This integration is first performed with respect to x , which extends from $x = 0$ to $x = 1 - y$, and then with respect to y , which extends from $y = 0$ to $y = 1$. Therefore,

$$\begin{aligned}\iint_R (x^2 + y^2) dx dy &= \int_0^1 \left(\int_0^{1-y} (x^2 + y^2) dx \right) dy = \int_0^1 \left[\frac{x^3}{3} + xy^2 \right]_0^{1-y} dy \\ &= \int_0^1 \left[\frac{(1-y)^3}{3} + (1-y)y^2 \right] dy \\ &= \left[\frac{(1-y)^4}{-12} + \frac{y^3}{3} - \frac{y^4}{4} \right]_0^1 = \frac{1}{3} - \frac{1}{4} + \frac{1}{12} = \frac{1}{6}.\end{aligned}$$



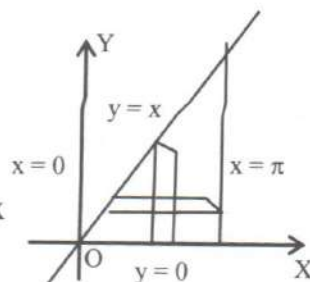
8. Evaluate $\int_0^\pi \int_0^x \sin y dy dx$

Solution: The integral is

$$\int_0^\pi \int_0^x \sin y dy dx = \int_0^\pi \left(\int_0^x \sin y dy \right) dx.$$

Here the region is bounded by $x = 0$, $x = \pi$, $y = 0$ and $y = x$ as shown in the figure. This is first performed with respect to y , which varies from $y = 0$ and $y = x$, and then with respect to x , which varies from $x = 0$ to $x = \pi$. Therefore,

$$\begin{aligned}\int_0^\pi \int_0^x \sin y dy dx &= - \int_0^\pi [\cos y]_0^x dx \\ &= - \int_0^\pi (\cos x - 1) dx = \int_0^\pi (1 - \cos x) dx \\ &= [x - \sin x]_0^\pi = \pi.\end{aligned}$$



9. Evaluate $\iint r^3 dr d\theta$ over the area included between the circles $r = 2 \sin \theta$ and $r = 4 \sin \theta$

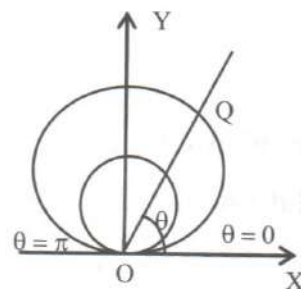
Solution: Here, the integral is given by

$$\iint r^3 dr d\theta,$$

where the region is between the circles $r = 2 \sin \theta$ and $r = 4 \sin \theta$ as shown in the figure. This integral is first integrated with respect to r , with r ranging from $r = 2 \sin \theta$ to $r = 4 \sin \theta$. To cover the entire region, θ varies from $\theta = 0$ to π . Therefore,

$$\begin{aligned}\iint r^3 dr d\theta &= \int_0^\pi \left(\int_{2 \sin \theta}^{4 \sin \theta} r^3 dr \right) d\theta = \int_0^\pi \left[\frac{r^4}{4} \right]_{2 \sin \theta}^{4 \sin \theta} d\theta \\ &= 60 \int_0^\pi \sin^4 \theta d\theta = 120 \int_0^{\pi/2} \sin^4 \theta d\theta.\end{aligned}$$

Applying the formula of Gamma function, we get



$$\iint r^3 dr d\theta = 120 \frac{\sqrt{\pi} \Gamma\left(\frac{4+1}{2}\right)}{2 \Gamma\left(\frac{4+2}{2}\right)} = 120 \frac{\sqrt{\pi} \frac{3}{2} \times \frac{1}{2} \sqrt{\pi}}{2 \times 2 \times 1} = \frac{45\pi}{2}.$$

III. Evaluate $\iint r \sin \theta dr d\theta$ over the area of the Cardioid $r = a(1 + \cos \theta)$ above the initial line

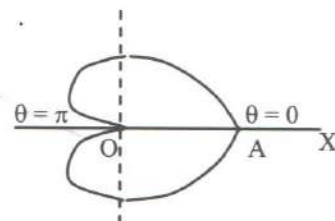
Solution: The integral is first integrated with respect to r , where the region covered by $r = 0$ to $r = a(1 + \cos \theta)$, and θ ranges from $\theta = 0$ to $\theta = \pi$, as shown in the figure. Therefore,

$$\begin{aligned} \iint r \sin \theta dr d\theta &= \int_0^\pi \left(\int_0^{a(1+\cos\theta)} r \sin \theta dr \right) d\theta = \int_0^\pi \sin \theta \left[\frac{r^2}{2} \right]_0^{a(1+\cos\theta)} d\theta \\ &= \frac{a^2}{2} \int_0^\pi \sin \theta (1 + \cos \theta)^2 d\theta. \end{aligned}$$

Putting $1 + \cos \theta = t$, so that $-\sin \theta d\theta = dt$ in this integral.

When $\theta = 0$, $t = 2$, and when $\theta = \pi$, $t = 0$, we get

$$\iint r \sin \theta dr d\theta = -\frac{a^2}{2} \int_2^0 t^2 dt = -\frac{a^2}{2} \left[\frac{t^3}{3} \right]_2^0 = -\frac{a^2}{2} \left[0 - \frac{8}{3} \right] = \frac{4a^2}{3}.$$



Exercise - 4

1. Evaluate the following integrals

(i) $\int_0^3 \int_1^2 xy(x+y) dx dy$

(iii) $\int_0^2 \int_0^{\sqrt{2x-x^2}} x dx dy$

(v) $\int_0^1 \int_0^{x^2} e^{y/x} dy dx$

(vii) $\int_0^1 \int_0^x e^{y/x} dy dx$

(ix) $\int_0^1 \int_y^{\sqrt{y}} (x^2 + y^2) dy dx$

(ii) $\int_0^3 \int_0^1 (x^2 + 3y^2) dy dx$

(iv) $\int_0^1 \int_x^{\sqrt{x}} (x^2 + y^2) dx dy$

(vi) $\int_0^a \int_0^{\sqrt{a^2-y^2}} dx dy$

(viii) $\int_1^{\log 8} \int_0^{\log y} e^{x+y} dx dy$

(x) $\int_0^a \int_0^{\sqrt{a^2-y^2}} \sqrt{a^2-x^2-y^2} dy dx$

2. Evaluate $\iint_R (x+y)^2 dx dy$ where R is the region bounded by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

3. Evaluate $\iint_R xy(x+y) dx dy$, and R is region over the area between the parabola $y = x^2$ and the line $y = x$

4. Evaluate $\iint_R xy \, dx \, dy$ where R is the region bounded by the x -axis, the ordinate $x = 2a$ and the curve $x^2 = 4ay$
5. Evaluate $\iint_R \sqrt{xy - y^2} \, dx \, dy$ where R is region bounded by the triangle with vertices $(0, 0)$, $(10, 1)$ and $(1, 1)$
6. Evaluate $\iint_R xy \, dx \, dy$ where R is the region over the area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ in the first quadrant
7. Evaluate $\iint_R x^2 \, dx \, dy$, where R is the region in the first quadrant bounded by the hyperbola $xy = 16$ and the line $y = x$, $y = 0$ and $x = 8$
8. Evaluate $\iint_R y \, dy \, dx$ where R is the region bounded by $x^2 + 2y^2 = 4$, the ordinate $y = 0$, $y = \sqrt{2}$
9. Evaluate $\iint_R r^3 \, dr \, d\theta$ over the area bounded between the circles $r = 2 \cos \theta$ and $r = 4 \cos \theta$
10. Evaluate $\iint_R r^2 \sin \theta \, dr \, d\theta$ where R is the semi-circle $r = 2a \cos \theta$ above the initial line
11. Evaluate the following integrals

(i) $\int_0^{\pi/2} \int_0^{\pi/2} \sin(\theta + \phi) \, d\theta \, d\phi$

(ii) $\int_0^{\pi} \int_0^{a \sin \theta} r \, dr \, d\theta$

(iii) $\int_0^{\pi} \int_0^{a\theta} r^3 \, dr \, d\theta$

(iv) $\iint_R \frac{r \, d\theta \, dr}{\sqrt{a^2 + r^2}}$ over one loop $r^2 = a^2 \cos 2\theta$

Answers

1. (i) 24 (ii) 12 (iii) $\frac{\pi}{2}$ (iv) $\frac{3}{35}$ (v) $\frac{1}{2}$ (vi) $\frac{\pi a^2}{4}$ (vii) $\frac{1}{2} (e - 1)$
- (viii) $8 \log 8 - 16 + e$ (ix) $\frac{3}{35}$ (x) $\frac{\pi a^3}{6}$ 2. $\frac{\pi ab}{4} (a^2 + b^2)$ 3. $\frac{3}{56}$ 4. $\frac{a^4}{3}$
5. 6 6. $\frac{a^2 b^2}{8}$ 7. 448 8. $\frac{8}{3}$ 9. $\frac{45\pi}{2}$ 10. $\frac{2a^3}{3}$
11. (i) 2 (ii) $\frac{\pi a^2}{4}$ (iii) $\frac{a^4 \pi^5}{20}$ (iv) $\left(2 - \frac{\pi}{2}\right) a$

2.1.3 Change of Order of Integration

In Calculus, the interchange of the order of integration is a technique used to transform iterated integrals into potentially simpler forms by changing the sequence in which integrations are performed. In some cases, the order of integration can be validly interchanged, while in others it cannot. Consider the iterated integral

$$\int_a^z dx \int_a^x h(y) \, dy.$$

In this expression, the inner integral is calculated first with respect to y while x is held constant. This

$$= \frac{1}{m} \int_0^1 x^{l-1} (1-x)^m dx = \frac{1}{m} \int_0^1 x^{l-1} (1-x)^{m+1-1} dx.$$

Recognizing both integrals as Beta functions, we get

$$\iint_R x^{l-1} y^{m-1} dx dy = \frac{1}{m} \beta(l, m+1).$$

By applying the relationships between the Beta and Gamma functions, we obtain

$$\iint_R x^{l-1} y^{m-1} dx dy = \frac{1}{m} \frac{\Gamma(m+1) \Gamma l}{\Gamma(m+1+l)}.$$

Using $\Gamma(m+1) = m \Gamma m$, we get

$$\iint_R x^{l-1} y^{m-1} dx dy = \frac{1}{m} \frac{m \Gamma m \Gamma l}{\Gamma(m+l+1)} = \frac{\Gamma l \Gamma m}{\Gamma(l+m+1)},$$

$$\therefore \iint_R x^{l-1} y^{m-1} dx dy = \frac{\Gamma l \Gamma m}{\Gamma(l+m+1)}.$$

Workout Examples

1. Evaluate $\int_0^1 \int_0^2 \int_1^2 x^2 yz dx dy dz$

Solution: We have

$$\begin{aligned} \int_0^1 \int_0^2 \int_1^2 x^2 yz dx dy dz &= \int_0^1 \int_0^2 \left(\int_1^2 z dz \right) x^2 y dx dy = \int_0^1 \int_0^2 x^2 y \left[\frac{z^2}{2} \right]_1^2 dx dy \\ &= \frac{3}{2} \int_0^1 x^2 \left(\int_0^2 y dy \right) dx = \frac{3}{2} \int_0^1 x^2 \left[\frac{y^2}{2} \right]_0^2 dx = \int_0^1 3x^2 dx = 1. \end{aligned}$$

2. Evaluate $\int_0^1 \int_{y^2}^1 \int_0^{1-x} x dz dx dy$

Solution: Here, the triple integral is given by

$$\begin{aligned} \int_0^1 \int_{y^2}^1 \int_0^{1-x} x dz dx dy &= \int_0^1 \int_{y^2}^1 x \left(\int_0^{1-x} dz \right) dx dy = \int_0^1 \int_{y^2}^1 x(1-x) dx dy \\ &= \int_0^1 \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_{y^2}^1 dy = \int_0^1 \left(\frac{1}{2} - \frac{1}{3} - \frac{y^4}{2} + \frac{y^6}{3} \right) dy \\ &= \left[\frac{y}{6} - \frac{y^5}{10} + \frac{y^7}{21} \right]_0^1 = \frac{1}{6} - \frac{1}{10} + \frac{1}{21} = \frac{4}{35}. \end{aligned}$$

3. Evaluate $\iiint_R (x+y+z) dx dy dz$ where $R: 0 \leq x \leq 1, 1 \leq y \leq 2, 2 \leq z \leq 3$

Solution: Here, the triple integral is given by

$$\begin{aligned} \iiint_R (x+y+z) dx dy dz &= \int_0^1 \int_1^2 \left(\int_2^3 (x+y+z) dz \right) dy dx \\ &= \int_0^1 \int_1^2 \left[zx + yz + \frac{z^2}{2} \right]_2^3 dy dx = \int_0^1 \left[\left(\int_1^2 x + y + \frac{5}{2} \right) dy \right] dx \end{aligned}$$

$$= \int_0^1 \left[xy + \frac{y^2}{2} + \frac{5y}{2} \right]_1^2 dx = \int_0^1 (x+4) dx = \left[\frac{x^2}{2} + 4x \right]_0^1 = \frac{9}{2}.$$

4. Evaluate $\int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} \frac{dz dy dx}{\sqrt{1-x^2-y^2-z^2}}$

Solution: Here, the triple integral is given by

$$\begin{aligned} \int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} \frac{dz dy dx}{\sqrt{1-x^2-y^2-z^2}} &= \int_0^1 \int_0^{\sqrt{1-x^2}} \left(\int_0^{\sqrt{1-x^2-y^2}} \frac{dz}{\sqrt{1-x^2-y^2-z^2}} \right) dy dx \\ &= \int_0^1 \int_0^{\sqrt{1-x^2}} \left[\sin^{-1} \frac{z}{\sqrt{1-x^2-y^2}} \right]_0^{\sqrt{1-x^2-y^2}} dy dx = \frac{\pi}{2} \int_0^1 \left(\int_0^{\sqrt{1-x^2}} dy \right) dx \\ &= \frac{\pi}{2} \int_0^1 \sqrt{1-x^2} dx = \frac{\pi}{2} \left[\frac{x\sqrt{1-x^2}}{2} + \frac{1}{2} \sin^{-1} x \right]_0^1 = \frac{\pi^2}{8}. \end{aligned}$$

5. Evaluate $\int_0^{\pi/2} \int_0^{\sin \theta} \int_0^{\frac{a^2-r^2}{a}} r dz dr d\theta$

Solution: Here, the triple integral is given by

$$\begin{aligned} \int_0^{\pi/2} \int_0^{\sin \theta} \int_0^{\frac{a^2-r^2}{a}} r dz dr d\theta &= \int_0^{\pi/2} \int_0^{\sin \theta} \left(\int_0^{\frac{a^2-r^2}{a}} r dz \right) dr d\theta \\ &= \int_0^{\pi/2} \left(\int_0^{\sin \theta} \frac{r(a^2-r^2)}{a} dr \right) d\theta = \frac{1}{a} \int_0^{\pi/2} \left[\frac{a^2 r^2}{2} - \frac{r^4}{4} \right]_0^{\sin \theta} d\theta \\ &= \frac{1}{a} \int_0^{\pi/2} \left(\frac{a^4 \sin^2 \theta}{2} - \frac{a^4 \sin^4 \theta}{4} \right) d\theta = \frac{a^3}{4} \int_0^{\pi/2} (2 \sin^2 \theta - \sin^4 \theta) d\theta \\ &= \frac{a^3}{4} \left[\frac{2\sqrt{\pi} \Gamma\left(\frac{2+1}{2}\right)}{2\Gamma\left(\frac{2+2}{2}\right)} - \frac{\sqrt{\pi} \Gamma\left(\frac{4+1}{2}\right)}{2\Gamma\left(\frac{4+2}{2}\right)} \right] = \frac{a^3}{4} \left[\sqrt{\pi} \frac{1}{2} \sqrt{\pi} - \frac{\sqrt{\pi} \frac{3}{2} \times \frac{1}{2} \sqrt{\pi}}{2 \times 2 \times 1} \right] \\ &= \frac{a^3}{4} \left(\frac{\pi}{2} - \frac{3\pi}{16} \right) = \frac{5\pi a^3}{64}. \end{aligned}$$

6. Evaluate $\int_0^{\log 2} \int_0^x \int_0^{x+\log y} e^{x+y+z} dz dy dx$

Solution: Here, the triple integral is given by

$$\begin{aligned} \int_0^{\log 2} \int_0^x \int_0^{x+\log y} e^{x+y+z} dz dy dx &= \int_0^{\log 2} \int_0^x \left(\int_0^{x+\log y} e^z dz \right) e^{x+y} dy dx \\ &= \int_0^{\log 2} \int_0^x [e^z]_0^{x+\log y} e^{x+y} dy dx = \int_0^{\log 2} \int_0^x (y e^x - 1) e^{x+y} dy dx \end{aligned}$$

$$\begin{aligned}
&= \int_0^{\log 2} \left(\int_0^x (y e^y e^x - e^y) dy \right) e^x dx = \int_0^{\log 2} [y e^{x+y} - e^{y+x} - e^y]_0^x e^x dx \\
&= \int_0^{\log 2} (x e^{3x} - e^{3x} + e^x) dx = \left[\frac{x e^{3x}}{3} - \frac{e^{3x}}{9} - \frac{e^{3x}}{3} + e^x \right]_0^{\log 2} \\
&= \frac{8}{3} \log 2 - \frac{8}{9} - \frac{8}{3} + 2 + \frac{1}{9} + \frac{1}{3} - 1 = \frac{8}{3} \log 2 - \frac{19}{9}.
\end{aligned}$$

7. Evaluate $\int_1^e \int_1^{\log y} \int_1^{e^x} \log z \, dz \, dx \, dy$

Solution: Here, the triple integral is given by

$$\begin{aligned}
\int_1^e \int_1^{\log y} \int_1^{e^x} \log z \, dz \, dx \, dy &= \int_1^e \int_1^{\log y} \left(\int_1^{e^x} \log z \, dz \right) dx \, dy = \int_1^e \int_1^{\log y} [z \log z - z]_1^{e^x} dx \, dy \\
&= \int_1^e \left(\int_1^{\log y} (x e^x - e^x + 1) dx \right) dy = \int_1^e [x e^x - e^x + x]_1^{\log y} dy \\
&= \int_1^e [(y \log y - e) - 2y + 2e + \log y - 1] dy = \int_1^e (y \log y - 2y + \log y + e - 1) dy \\
&= \left[\frac{y^2}{2} \log y - \frac{y^2}{4} - y^2 + y \log y - y + e y - y \right]_1^e = \frac{e^2}{2} - \frac{5}{4}(e^2 - 1) + e - 2(e - 1) + e(e - 1) \\
&= \frac{e^2}{2} - \frac{5e^2}{4} + \frac{5}{4} + e - 2e + 2 + e^2 - e = \frac{e^2}{4} - 2e + \frac{13}{4} = \frac{1}{4}(e^2 - 8e + 13)
\end{aligned}$$

8. Evaluate $\int_0^a \int_0^x \int_0^{x+y} e^{x+y+z} \, dx \, dy \, dz$

Solution: Here, the triple integral is given by

$$\begin{aligned}
\int_0^a \int_0^x \int_0^{x+y} e^{x+y+z} \, dx \, dy \, dz &= \int_0^a \int_0^x \left(\int_0^{x+y} e^z \, dz \right) e^{x+y} dy \, dx = \int_0^a \left(\int_0^x (e^{x+y} - 1) e^y dy \right) e^x dx \\
&= \int_0^a \left[\frac{e^x e^{2y}}{2} - e^y \right]_0^x e^x dx = \int_0^a \left(\frac{e^{3x}}{2} - e^x - \frac{e^x}{2} + 1 \right) e^x dx = \int_0^a \left(\frac{e^{4x}}{2} - \frac{3e^{2x}}{2} + e^x \right) dx \\
&= \left[\frac{e^{4x}}{8} - \frac{3e^{2x}}{4} + e^x \right]_0^a = \frac{e^{4a}}{8} - \frac{3e^{2a}}{4} + e^a - \frac{1}{8} + \frac{3}{4} - 1 = \frac{1}{8}(e^{4a} - 6e^{2a} + 8e^a - 3).
\end{aligned}$$

9. Evaluate $\iiint_R (2x + y) \, dx \, dy \, dz$ where R is closed region bounded by the cylinder $z = 4 - x^2$ and the planes $x = 0$, $y = 0$, $y = 2$, and $z = 0$

Solution: Here, the triple integral is given by

$$\iiint_R (2x + y) \, dx \, dy \, dz,$$

where the region is bounded by the planes $z = 0$ and the surface $z = 4 - x^2$ and the planes $x = 0$, $y = 0$, $y = 2$ as shown in the figure. The cylinder $z = 4 - x^2$ intersects the x -axis at $(2, 0, 0)$ and the y -axis at $(0, 2, 0)$ in the given region. Therefore, the limits for z are from $z = 0$ to $z = 4 - x^2$, for y from $y = 0$ to $y = 2$, and for x from $x = 0$ to $x = 2$. So, the integral can be written as

$$\iiint_R (2x + y) \, dx \, dy \, dz = \int_0^2 \int_0^2 \left(\int_0^{4-x^2} (2x + y) \, dz \right) dy \, dx = \int_0^2 \int_0^2 (4 - x^2)(2x + y) \, dy \, dx$$